A/B Test Analysis:

Subscription Pricing Experiment

1. Experiment Overview

The experiment compared two subscription offers designed to evaluate whether highlighting a 50% discount would increase the conversion rate from app installation to payment.

- **Group A (Control): Users were offered a premium subscription for $4.99.**
- **Group B (Variant): Users were offered the same $4.99 subscription, with messaging emphasizing a 50% discount.**

The key difference between the two groups was the addition of the 50% discount message in Group B.

Experiment period

- **Start date: July 3, 2023**
- **End date: July 25, 2023**
- **Duration: 21 days**

2. Users and Conversions

Metric	Group A (Control)	Group B (Variant)
Number of users	10,013	9,985
Number of conversions	611	889
Conversion rate	6.10%	8.90%

Group B achieved a higher conversion rate than Group A. The observed absolute difference was approximately 2.80 percentage points, indicating a potentially meaningful improvement in subscription conversion.

3. Statistical Analysis

Two-Proportion Z-Test

A two-proportion Z-test was used to compare conversion rates between the two independent groups. This test is appropriate for evaluating differences in proportions when sample sizes are sufficiently large.

Hypotheses

- **Null hypothesis (H_0): The conversion rates in Groups A and B are equal.**
- **Alternative hypothesis (H_1): The conversion rates in Groups A and B are different.**

Test results

- **Z-statistic: -7.5197**
- **P-value: approximately 5.49×10^{-14}**
- **Significance level (α): 0.05**

Statistical Conclusion

The p-value is substantially lower than the significance level of 0.05. Therefore, the null hypothesis is rejected.

The experiment provides strong statistical evidence of a difference in conversion rates between the two groups. Group B achieved a higher observed conversion rate than Group A.

4. Key Findings

- **Group B achieved a conversion rate of 8.90%, compared with 6.10% in Group A.**
- **The absolute conversion-rate difference was approximately 2.80 percentage points.**
- **The difference was statistically significant at the 5% significance level.**
- **Highlighting the 50% discount was associated with a higher conversion rate in this experiment.**

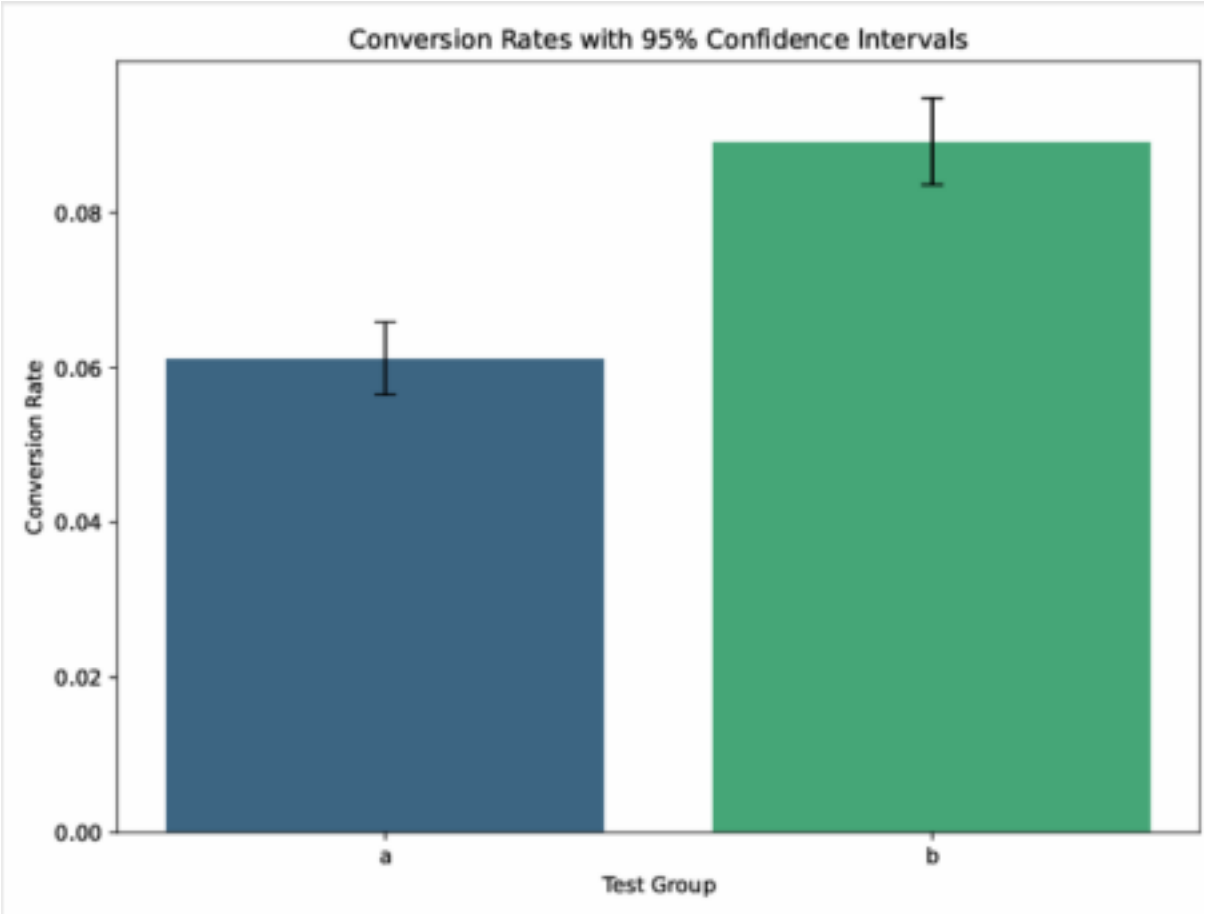
5. Business Recommendation

Based on the observed results, the discount-focused subscription offer in Group B is the preferred option for further consideration.

Before implementing the change broadly, the business should also evaluate its impact on revenue, profit margins, customer retention, and the long-term value of subscribers. A higher conversion rate alone does not necessarily guarantee higher overall profitability.

Final conclusion: Group B significantly outperformed Group A in conversion rate during the experiment. The results support further consideration of the

discount-focused messaging, subject to business and profitability validation.



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